

Krawtchouk functions maximize every even moment on a Boolean Fourier level

Automated mathematical discovery run `fa_banach_001`

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Status. Candidate full solution, likely valid, subject to human review. For every even exponent $p = 2m$, the two maxima in the source’s appendix Question B.2(6) are attained by a permutation-invariant Krawtchouk function. The $p = 4$ case was proved by Aaronson in 2018; the argument below extends the compression principle to all even moments.

1 Source question and provenance

The source is Y. Polyanskiy, *Hypercontractivity of spherical averages in Hamming space*, arXiv:1309.3014 [1]. Write $G = \mathbb{F}_2^n$, equip it with uniform probability measure, and let Π_a denote orthogonal projection onto the Fourier level

$$\mathcal{H}_a = \text{span}\{\chi_A : A \subseteq [n], |A| = a\}, \quad \chi_A(x) = (-1)^{\sum_{i \in A} x_i}.$$

The source asks whether, for $p = 2m$, the maxima of

$$\frac{\|f_a\|_p}{\|f_a\|_2} \quad \text{and} \quad \frac{\|f_a\|_p}{\|f\|_2}, \quad f_a = \Pi_a f,$$

are achieved by permutation-invariant functions. The exact passage is reproduced below.

B.2. Open questions. TODO:

1. Semenov-Shneiberg: relate to averages of symmetric functions over codes with bounded mindist/maxdist.
2. $\|\Pi_a\|_{p \rightarrow q}$ via contour integration, Minkowski and Bonami-Gross.
3. More generally, identities and properties of $BSC(z)$ complex-noise channels.
4. Implications in terms of additive energy $E(A, B)$ when B is permutation invariant?
5. Perhaps prove the following general property of groups:

$$\sup_x \left| \frac{K_j(x)}{K_j(0)} \right| \cdot \|(\cdot) * 2^{-n} K_j\|_{p(j) \rightarrow 2} = O(1),$$

where $p(j) = 1 + \sigma_2^2$ determined by the spectral gap?

6. Study whether maxima of $\frac{\|f_a\|_p}{\|f_a\|_2}$ and $\frac{\|f_a\|_p}{\|f\|_2}$ is achieved by permutation invariant functions when $p = 2m$. (Not true for $a = 1$ and $2 < p < 3$, but maybe for $p = 2m$?).

Provenance note. This question occurs in the arXiv source tree inside an appendix guarded by the source’s `ifmapx` switch; it is not printed in the official 24-page arXiv PDF. The crop is from page 27 of a DVI/PDF build of the original source with its documented appendix job name `hc_hamming_apx`. The official PDF is retained as `source_paper.pdf`.

2 Result and proof intuition

Define the degree- a Krawtchouk function

$$K_a(x) = \sum_{|A|=a} \chi_A(x), \quad \|K_a\|_2 = \binom{n}{a}^{1/2}.$$

Theorem 1 (Exact even-moment level norm). *For all $n \geq 1$, $0 \leq a \leq n$, and integers $m \geq 1$,*

$$\sup_{0 \neq g \in \mathcal{H}_a} \frac{\|g\|_{2m}}{\|g\|_2} = \frac{\|K_a\|_{2m}}{\|K_a\|_2}. \quad (1)$$

The supremum may be taken over real- or complex-valued functions. Moreover,

$$\|\Pi_a\|_{L_2(G) \rightarrow L_{2m}(G)} = \frac{\|K_a\|_{2m}}{\|K_a\|_2}. \quad (2)$$

Thus both maxima in the source question are attained by the permutation-invariant function K_a .

The proof works on the Fourier-coefficient sphere $S(n, a) = \{A \subseteq [n] : |A| = a\}$. First discard coefficient phases: the even-moment expansion can only increase when every coefficient is replaced by its modulus. Next choose two coordinates i, j . Whenever two coefficients sit at the endpoints of a Johnson-graph edge $A \leftrightarrow A \triangle \{i, j\}$, replace both by their root-mean-square. This preserves the coefficient ℓ_2 norm.

For the fourth moment, monotonicity of this compression is Aaronson's key observation [2]. At a general $2m$ -th moment, fix all coordinates outside i, j in the $2m$ -tuple expansion. The remaining sum is an even- or odd-parity tensor. Its elementary sharp product bound shows that the same root-mean-square compression is monotone for every even moment. Repeating the compressions averages the squared coefficients over the connected Johnson graph, leaving the constant vector and hence K_a .

3 The parity-tensor inequality

Lemma 2 (Parity block bound). *Let $r \geq 2$, let $u_\ell, v_\ell \geq 0$, and let $\epsilon \in \{0, 1\}$. Then*

$$\sum_{\substack{\sigma \in \{0,1\}^r \\ \sum_\ell \sigma_\ell \equiv \epsilon \pmod{2}}} \prod_{\ell=1}^r (u_\ell \mathbf{1}_{\{\sigma_\ell=0\}} + v_\ell \mathbf{1}_{\{\sigma_\ell=1\}}) \leq 2^{r/2-1} \prod_{\ell=1}^r (u_\ell^2 + v_\ell^2)^{1/2}. \quad (3)$$

The right side is exactly the left side after every pair (u_ℓ, v_ℓ) is replaced by (ρ_ℓ, ρ_ℓ) , where $\rho_\ell = ((u_\ell^2 + v_\ell^2)/2)^{1/2}$.

Proof. The parity filter writes the left side as

$$\frac{1}{2} \left\{ \prod_\ell (u_\ell + v_\ell) + (-1)^\epsilon \prod_\ell (u_\ell - v_\ell) \right\}.$$

Zero pairs may be removed, so normalize $t_\ell = (u_\ell^2 + v_\ell^2)^{1/2} > 0$, and set

$$x_\ell = \frac{u_\ell + v_\ell}{\sqrt{2} t_\ell}, \quad y_\ell = \frac{u_\ell - v_\ell}{\sqrt{2} t_\ell}.$$

Then $x_\ell^2 + y_\ell^2 = 1$. Since $r \geq 2$,

$$\prod_\ell |x_\ell| + \prod_\ell |y_\ell| \leq |x_1 x_2| + |y_1 y_2| \leq \sqrt{(x_1^2 + y_1^2)(x_2^2 + y_2^2)} = 1.$$

Substitution proves (3). After replacement there are 2^{r-1} admissible parity strings, each contributing $2^{-r/2} \prod t_\ell$, proving the final assertion. $\square$

4 Proof of the theorem

Proof of Theorem 1. Write $g = \sum_{|A|=a} c_A \chi_A$. Expanding against the orthonormal characters gives

$$\|g\|_{2m}^{2m} = \sum_{\substack{A_1, \dots, A_m, B_1, \dots, B_m \in S(n, a) \\ A_1 \triangle \dots \triangle A_m = B_1 \triangle \dots \triangle B_m}} c_{A_1} \cdots c_{A_m} \overline{c_{B_1} \cdots c_{B_m}}. \quad (4)$$

Taking absolute values term by term shows that this is at most the same moment with c_A replaced by $|c_A|$, while Parseval leaves $\|g\|_2^2 = \sum_A |c_A|^2$ unchanged. It therefore suffices to assume all coefficients are nonnegative.

Fix $i < j$. If A contains exactly one of i, j , put $A' = A \triangle \{i, j\}$ and replace both $c_A, c_{A'}$ by

$$\rho_A = \left(\frac{c_A^2 + c_{A'}^2}{2} \right)^{1/2}.$$

Leave the other coefficients fixed. This compression preserves $\sum_A c_A^2$.

To compare moments, use the equivalent $2m$ -tuple form of (4), whose constraint is $A_1 \triangle \dots \triangle A_{2m} = \emptyset$. Group terms after deleting coordinates i, j from every A_ℓ . If r of the resulting bases have size $a - 1$, their lifts have coordinate patterns 10 or 01. The other lifts, of sizes a and $a - 2$, are fixed as 00 and 11. A block contributes only when r is even. For $r = 0$ it is unchanged. For even $r \geq 2$, the two coordinate parity conditions reduce to one prescribed parity of the r binary choices. After removing the common product of fixed coefficients, the block is the left side of (3). Lemma 2 says precisely that the compression cannot decrease it. Hence every coordinate-pair compression preserves $\|g\|_2$ and increases $\|g\|_{2m}$.

On squared coefficients $w_A = c_A^2$, this operation is ordinary averaging across every Johnson-graph edge carrying the label i, j . Cycle through all coordinate pairs and repeat. These doubly stochastic averaging maps have a positive diagonal, and the union of their edges is the connected Johnson graph $J(n, a)$. The standard finite consensus argument therefore gives

$$w_A \longrightarrow \binom{n}{a}^{-1} \sum_{|B|=a} w_B \quad \text{for every } A \in S(n, a).$$

Continuity of the finite moment polynomial shows that the original ratio is at most the ratio for constant nonnegative coefficients. Their Fourier extension is a scalar multiple of K_a , proving (1).

Finally, for arbitrary f , orthogonality gives

$$\|\Pi_a f\|_{2m} \leq \frac{\|K_a\|_{2m}}{\|K_a\|_2} \|\Pi_a f\|_2 \leq \frac{\|K_a\|_{2m}}{\|K_a\|_2} \|f\|_2.$$

Equality is attained at $f = K_a$, which proves (2) and the second assertion in the source question. $\square$

5 Verification, literature boundary, and limitations

The proof was checked in three independent ways. First, direct numerical optimization over every tested level through $n = 7$, for moments $p = 4, 6, 8$, returned the constant-magnitude coefficient orbit. Second, 20,000 random pair-compression tests in each of nine configurations through $p = 10$ found no decrease. Third, the block formula was checked directly against enumeration of parity strings. These computations are sanity checks only; the proof is the exact parity-tensor argument above.

The case $p = 4$ is not new: Aaronson proved exact constant-coefficient extremality for additive energy [2], answering a formulation of Kirshner–Samorodnitsky [3]. Kirshner–Samorodnitsky

later obtained general- p moment bounds that are asymptotically tight up to lower-order factors [4]; that work does not state the exact all-even-moment identity above. Bounded searches through 13 August 2026 used the exact source wording, “Krawtchouk extremizer”, “even moments”, higher additive energy, and the citation neighborhood of arXiv:1805.05295. No exact all-even- p theorem was found. Novelty confidence is therefore moderate, not definitive.

The theorem addresses only the even exponents explicitly proposed in the source. It makes no claim for non-even $p > 2$, where the polynomial expansion and parity compression are unavailable; the source already notes failure of the analogous symmetry assertion at degree one for $2 < p < 3$. It also does not classify all equality cases.

References

- [1] Y. Polyanskiy, *Hypercontractivity of spherical averages in Hamming space*, arXiv:1309.3014.
- [2] J. Aaronson, *Functions with large additive energy supported on a Hamming sphere*, arXiv:1805.05295, 2018.
- [3] N. Kirshner and A. Samorodnitsky, *On the $\ell_4 : \ell_2$ ratio of functions with restricted Fourier support*, arXiv:1801.08507.
- [4] N. Kirshner and A. Samorodnitsky, *A moment ratio bound for polynomials and some extremal properties of Krawtchouk polynomials and Hamming spheres*, arXiv:1909.11929.